

RAHUL J HIRUR

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OBJECTIVE

Computational Engineer specializing in Applied AI and Foundation Models, with proven expertise in integrating LLMs and Vision pipelines into scalable industrial systems, looking for roles in the area of Applied AI.

EDUCATION

Master of Science (M. Sc.) in Computational Engineering

Oct 2022 – Mar 2026

Friedrich Alexander Universität Erlangen-Nürnberg

Relevant Coursework: Deep Learning, Computer Vision, Pattern Recognition, Image Video and Multi-Dimensional Signal Processing.

Master of Science (M. Sc.) in Computational Science

Sep 2023 – Mar 2026

Università della Svizzera italiana, Switzerland

Relevant Coursework: Data Analytics, Scientific Learning, Efficient Computational Algorithm, Robotics.

Double Degree Program – Erasmus

Bachelor in Engineering: Major in Mech. Engineering and minor in Computer Science

Aug 2015 – Aug 2019

KLE Technological University, India

SKILLS

Programming Languages

Python, Matlab, ROS, SQL, C

Deep Learning

PyTorch, TensorFlow, Hugging Face, Scikit-Learn, Open-CV

Web

Streamlit, Dash, Plotly, Power BI

CI-CD

Git, Jenkins, Kubernetes, Docker

Others

Siemens NX, NXOpen, Solidworks, SDLC, HiTL, Digital Signal Processing

EXPERIENCE

Masterarbeit - AI based High-Precision 3D Reconstruction and Perception

Feb 2025 – Dec 2025

FAPS – Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik

Erlangen, Germany

- Developed a complete end-to-end 3D reconstruction Generative (GenAI) pipeline, encompassing image acquisition, processing, and final 3D construction for real-time robotic perception.
- Investigated and implemented two distinct depth methodologies: a Stereo Foundation Model (e.g., NVIDIA FoundationStereo) and a Monocular Foundation Model (e.g., DepthAnything) for disparity and depth calculation.
- Designed and implemented Phase Shifting Profilometry (PSP) based Prompt Depth Anything model for accurate depth calculation.
- Applied advanced computer vision techniques for Depth Segmentation and geometric processing to ensure the generation of accurate, high-density 3D point clouds.

Werkstudent – Data Engineering and Analytics

Apr 2023 – Nov 2025

Robert Bosch GmbH - Vehicle Motion

Abstatt, Germany

- Developed metrics and algorithms to evaluate Electronic Stability Program (ESP) simulation results.
- Designed and implemented a variational autoencoder (VAE) to identify anomalies in simulated data.
- Built and deployed a Plausibility Check dashboard using Streamlit to assess ESP test bench results, utilizing Docker and Kubernetes for scalable container orchestration.
- Developed a Python-based Project Assessment tool to evaluate ESP project feasibility based on fatigue and environment conditions; managed deployment via Rancher.
- Migrated the ESP SPoC Tool from MATLAB to Python, streamlining the codebase.

System Interface Engineer

Robert Bosch - BGSW

Jan 2020 - Sep 2022

Bengaluru, India

- Analyzed and validated performance and fatigue loads for Brake Modulation Systems (ESP and IPB).
- Developed MATLAB based ETL pipelines to validate ESP simulation data, reducing manual analysis time by 95%.
- Developed a MobileNet-based CNN pipeline to analyze 3D CAD drawings and automatically detect and document mechanical design changes.
- Developed and maintained MATLAB-based visualization tools to monitor valve pressure modulations and detect mechanical failures.
- Engineered Jenkins pipelines to automate the ESP simulation workflow, increasing testing efficiency.

PROJECTS

Dhan-UI: Multimodal Hands-Free Surgical Robot Interface

- Implemented a vision pipeline using MediaPipe to track and interpret hand gestures, converting physical movements into string-based context for the Orchestrator LLM.
- Integrated Google Speech API to transcribe real-time voice commands, providing a parallel speech-to-text input stream to ensure redundancy in hands-free operation.
- Designed a central Orchestrator LLM using few-shot prompting to synthesize dual-channel inputs into structured, validated micro instruction commands.

Transformers Solving the Traveling Salesman Problem

- Developed a custom Transformer model to solve the traveling salesman problem.
- Testing the model versus the conventional methods, like the Greedy approach.
- Tuning the model and hyper parametric study.

Robotic Environment Identification and Obstacle avoidance

- Developed custom MobileNetV2 CNN model for real-time environment identification for MyT robot.
- Testing various models like ResNet, YOLO for faster and leaner identification.
- Implemented obstacle avoidance-based random walk algorithm for robot motion planning in ROS2.
- Built multiple environments in CoppeliaSim for robot testing and simulations.

HONOURS AND AWARDS

Bosch Automation Hackathon - First Prize

Jan 2021

Won the Bosch Automation Hackathon for the presentation of a computer vision based CAD drawing analyzer and report generator

Bosch Global Software Technologies

Bravo Award

Dec 2021

Presented for high effort, skillful contribution towards the ESP project

Bosch Global Software Technologies

LANGUAGE SKILLS

Mother tongue(s): Kannada, Hindi

Other language(s): English: Advanced Fluency(C1), German: Elementary Fluency(A2)

INTERESTS

Geopolitics, Running, Scripting, Audio-books, Technical Exploration